

Analysis of Top 5 H1 Persistent Loops

Summary Table

Rank	Persistence	Birth	Death	Process Count	Representative Processes
1	0.330	1.148	1.478	27	Ara operon, SOS Response, Quorum Sensing, Stringent Response, Catabolite Repression, Pho Regulon, Biofilm Formation
2	0.308	1.537	1.845	4	Aerobic Respiration, Cell Wall Integrity, DNA Replication, MAPK Mating (all yeast)
3	0.302	1.114	1.416	4	Lac Operon , Antibiotic Efflux Pumps, Phosphate Regulation, Translation Termination
4	0.283	1.186	1.469	17	Ara operon, Catabolite Repression, Stringent Response, Pho Regulon, Quorum Sensing, GAL Regulation, ER Stress
5	0.231	1.293	1.524	3	Two-Component Signaling (EnvZ-OmpR) , Oxidative Stress Response, Yeast ER-Associated Degradation

Detailed Analysis

H1 Loop #1 (Persistence = 0.330)

Contributing Processes (27 processes):

E. coli: Ara operon, Arginine biosynthesis, Base excision repair, Biofilm formation, Catabolite repression, Stringent response, Fatty acid synthesis, Heat shock, Maltose regulon, Nitrogen assimilation, Outer membrane assembly, Pho regulon, Quorum sensing, SOS response, Stringent response (ppGpp)

S. cerevisiae: Autophagy, ER stress response, GAL regulation, MAPK mating, Nitrogen metabolism, RNA splicing, Ubiquitin-proteasome, Vesicle trafficking, Cell polarity, ER-associated degradation, Vacuolar protein sorting

Bacillus subtilis: Competence development

Shared Biological Features: - Organism distribution: 15 E. coli, 11 S. cerevisiae, 1 Bacillus subtilis - Feedback structure: Many are known feedback/regulatory circuits (ara operon, SOS, stringent response, catabolite repression, Pho regulon, quorum sensing, heat shock, GAL, MAPK mating) - Circuit types: Gene regulation, metabolic regulation, stress response, signal transduction - Complexity: Avg nodes ~65, broad range of gate counts

Biological Interpretation: This large loop spans all three organisms and captures regulatory circuits with feedback: nutrient sensing (Pho, nitrogen, carbon), stress responses (SOS, heat shock, ER stress), and developmental signaling (quorum sensing, MAPK mating). The topology groups processes that share regulatory architecture rather than metabolic function.

H1 Loop #2 (Persistence = 0.308)

Contributing Processes (4 processes):

All S. cerevisiae: Aerobic respiration, Cell wall integrity (CWI), DNA replication, MAPK pheromone mating

Shared Biological Features: - Organism: 100% yeast - Feedback structure: MAPK mating and CWI are signaling pathways with feedback; DNA replication has checkpoint feedback - Circuit types: Metabolic pathway, stress response, DNA replication, signal transduction

Biological Interpretation: A tight yeast-specific loop. These processes share high node/conditional counts and logic gate structure. MAPK mating and CWI are both MAPK cascade pathways with feedback; aerobic respiration and DNA replication are core cellular processes with regulatory checkpoints.

H1 Loop #3 (Persistence = 0.302)

Contributing Processes (4 processes):

All E. coli: **Lac Operon**, Antibiotic efflux pumps, Phosphate regulation, Translation termination

Shared Biological Features: - Organism: 100% E. coli - Feedback structure: Lac operon is the canonical feedback circuit; Pho regulon has two-component feedback - Circuit types: Gene regulation, biological process, protein synthesis

Biological Interpretation: Lac operon appears in a top H1 loop. This loop groups processes with similar structural complexity (nodes, conditionals, gates). Lac operon's classic negative feedback may be topologically similar to other regulatory circuits; phosphate regulation (Pho) also uses two-component feedback. The inclusion of translation termination suggests shared graph-theoretic structure (branching, logic) rather than purely regulatory function.

H1 Loop #4 (Persistence = 0.283)

Contributing Processes (17 processes):

Overlap with Loop #1: Ara operon, Arginine biosynthesis, Catabolite repression, Stringent response, Fatty acid synthesis, Heat shock, Nitrogen assimilation, Pho regulon, Quorum sensing, Stringent (ppGpp); yeast ER stress, GAL, Nitrogen metabolism, Cell polarity, ERAD, Vacuolar sorting; Bacillus competence.

Shared Biological Features: - Organism: Mix of E. coli, yeast, Bacillus - Feedback structure: Strong overlap with Loop #1; many nutrient/stress regulatory circuits - Circuit types: Metabolic regulation, stress response, gene regulation

Biological Interpretation: A nested structure: Loop #4 is a subset of the processes in Loop #1, appearing at a slightly different birth scale. This suggests a hierarchy: the most persistent loop (#1) encompasses a broader regulatory circuit cloud; Loop #4 isolates a coherent subcluster (nutrient sensing + stress + developmental).

H1 Loop #5 (Persistence = 0.231)

Contributing Processes (3 processes):

E. coli: **Two-Component Signal Transduction (EnvZ-OmpR)**, Oxidative stress response

S. cerevisiae: ER-associated degradation

Shared Biological Features: - Organism: 2 E. coli, 1 yeast - Feedback structure: EnvZ-OmpR is the paradigm two-component system with feedback; OxyR/SoxRS oxidative stress has positive feedback - Circuit types: Signal transduction, stress response, protein quality control

Biological Interpretation: Two-component signaling (another classic feedback system) appears in a top H1 loop. This loop bridges organism and function: EnvZ-OmpR (osmoregulation) and oxidative stress (OxyR) share two-component

or sensory-regulatory logic; yeast ERAD has feedback (UPR). The topology captures cross-organism similarity in regulatory structure.

Key Findings

1. Are the loops biologically coherent?

Yes. Loops #1, #3, #4, and #5 prominently include known feedback circuits: lac operon, ara operon, two-component (EnvZ-OmpR), SOS, stringent response, catabolite repression, Pho regulon, quorum sensing, MAPK signaling. Loop #2 groups yeast processes with shared MAPK/checkpoint architecture.

2. Do feedback circuits cluster together?

Yes. Lac operon (Loop #3) and two-component signaling (Loop #5) appear in small, focused loops. The large Loop #1 aggregates many feedback circuits (SOS, stringent, Pho, quorum sensing, heat shock, GAL). The topology distinguishes “classic” single-circuit feedback (lac, EnvZ-OmpR) from “clouds” of regulatory processes.

3. Do organisms separate topologically?

Partially. Loop #2 is yeast-only; Loop #3 is E. coli-only. Loops #1, #4, #5 mix organisms. Organism-specific loops may reflect shared circuit architecture within a species; mixed loops suggest cross-species regulatory motifs.

4. What does this tell us?

The persistence diagram is capturing biological structure: feedback and regulatory circuits cluster in H1. The appearance of lac operon and two-component signaling in top loops (both textbook feedback systems) supports the hypothesis that TDA on structural features (nodes, gates) reflects regulatory logic. Mapper or persistent cohomology could further resolve circuit classes.